

Roll No.

TCE-604

**B. TECH. (CIVIL ENGINEERING)
(SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
QUANTITY ESTIMATION AND COSTING**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain and elaborate cubical content rate estimate and, preliminary estimate. (CO1)

OR

- (b) Explain and elaborate measurement sheet and abstract sheet. (CO1)
2. (a) Explain and elaborate cubical content rate estimate and, preliminary estimate. (CO1)

OR

- (b) Prepare the rough estimate for a proposed commercial complex for a municipal corporation for the following data : (CO1)

Plinth Area = 900 m²/floor

Ht. of each storey = 3.0 m

No. of storeys = G + 4

Cubical content rate = ₹ 1,175/m³

P. T. O.

(2)

Provided for the following as a percentage of structured cost :

- (i) Water supply and Sanitary arrangement – 7.5%
- (ii) Electrification – 6.5%
- (iii) Fluctuation of rates – 1.5%
- (iv) Contractors profit – 10%
- (v) Petty supervision and contingencies – 2.5%

3. (a) Explain the data required for the analysis of rates. (CO2)

OR

- (b) Explain the following terms : (CO2)

- (i) Cost of materials
- (ii) BSR
- (iii) Cost of labour
- (iv) Contractor's profit

4. (a) Analyse the rate for first class brickwork in superstructure with $20 \times 10 \times 10$ cm with (1 : 6) cement mortar considered 10 cum. Adopt suitable rates. (CO2)

OR

- (b) Explain overhead costs. (CO2)

5. (a) Calculate the quantities of material required for the following items of work : (CO2)

- (i) Cement plaster 20 mm thick cement mortar 1 : 3 for 400 m²
- (ii) Cement concrete 1 : 3 : 6 for 10 cum

OR

- (b) Explain task or out turn work and its effect on construction work. (CO2)

Roll No.

TCE-601

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ENVIRONMENTAL ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the relative merits of the separate systems of sewage and give the favorable site condition for the adoption of it. (CO1)

OR

- (b) A certain district of a city has a projected population of 80,000 residing over an area of 60 hectares. Find the design discharge for the sewer line, for the following data : (CO1)

(i) Rate of water supply = 220 L/d/person

(ii) Average impermeability coefficient for the entire area = 0.4

(iii) Time of concentration = 40 min.

The sewer line is to be designed for a flow equivalent to the wet weather flow (W.W.F.) plus twice the dry weather flow (D.W.F.). Assume that 80% of water supply reaches in sewer as wastewater.

P. T. O.

lib
24-4-2023
10-11-30

(2)

2. (a) Define and explain the following terms : (CO1)

(i) Inlet time and channel flow time

(ii) Critical rainfall duration and critical rainfall intensity.

OR

(b) List out the various source of sewage generation in a city. (CO1)

3. (a) list out various legislative requirement as per NGT for STP. (CO1)

OR

(b) Write a short note on BOD and COD for STP. (CO1)

4. (a) Write a short note of one pipe and two pipe plumbing system. (CO2)

OR

(b) Write short notes on the following : (CO2)

(i) Self-cleansing velocity

(ii) Non-scouring velocity

5. (a) Derive the expression of BOD to calculate the BOD at any time 't'.

(CO2)

OR

(b) The BOD_5 of a wastewater is 150 mg/L at 20°C. The k value is known to be 0.23 per day. What would be BOD_5 if the test was run at 15 °C ?

(CO2)

Roll No.

TCE-605

**B. TECH. (CE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
TRANSPORTATION ENGINEERING—I**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Describe the history of road construction. (CO1)

OR

- (b) Calculate the stopping sight distance on a national highway in rolling terrain at an ascending gradient of 3%. Assume design speed of 80 kmph, coefficient of friction as 0.35 and total reaction time of driver as 2.5 seconds. (CO1)

2. (a) Explain the factors controlling the highway alignment. (CO1)

OR

- (b) Calculate the length of transition curve and the shift using the following data : (CO1)

Design speed = 65 kmph

Radius of circular curve = 220 m

Allowable rate of introduction of super elevation (pavement rotated about the centre line) = 1 in 150

Pavement width including extra widening = 7.5 m

P. T. O.

(2)

3. (a) Calculate the maximum permissible speed on a horizontal curve of radius 125 m of a highway designed for a speed of 65 kmph to carry mixed traffic, if the super elevation is not to exceed 7 %. (CO1)

OR

- (b) Derive an expression for calculating the overtaking sight distance on a highway. (CO1)
4. (a) List different tests on road aggregates and mention their advantages and limitations. (CO2)

OR

- (b) What are the desirable properties of aggregate ? (CO2)
5. (a) List different tests on bitumen and mention the principle and uses of each test. (CO2)

OR

- (b) What are the desirable properties of subgrade soil ? (CO2)

Roll No.

TCE-603

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

WATER RESOURCES ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Enumerate different methods of stability analysis of gravity dam and explain in brief any *one* of them. (CO1)

OR

- (b) A gravity dam has the following dimension : (CO1)

Height of the dam = 90 m

Freeboard = 3 m

u/s slope = 0.1 H : 1 V

top width of dam = 8 m

Taking $K_h = 0.1$

Determine hydrodynamic earthquake pressure force and its moment at the base of dam.

P. T. O.

2. (a) How does the practical profile of a low gravity dam differ from that of the theoretical one and why ? (CO1)

OR

- (b) Design the practical profile of a gravity dam of stone masonry considering full uplift, from given the following data : (CO1)

R.L. of base of the dam = 1550 m

R. L. of F. R.L. = 1590.50 m

Specific gravity of the masonry = 2.40

Height of waves = 2.0 m.

Safe compressive stress for masonry = 1200 kN/m².

3. (a) What is meant by phreatic line ? How will you determine the phreatic line in case of homogeneous earthen dam ? (CO2)

OR

- (b) Determine the factor of safety of downstream slope of an earthen dam (homogeneous section) drawn to a scale of 1 cm = 4 m : (CO2)

Length of slip circle arc = 12 cm

Total area of N-rectangle = 15.5 cm²

Total area of T-rectangle = 7.0 cm²

Total area of U-rectangle = 5 cm²

Angle, $\phi = 26^\circ$

Cohesion, $c = 20$ kN/m²

Specific weight of soil 18 kN/m³

(CO2)

4. (a) Explain the thin cylinder theory of design of an arch dam. What are the limitations of this theory ? (CO2)

OR

- (b) Calculate the discharge per meter length of the earthen dam from the following data : (CO2)

Height of dam = 26 m.

Free board = 2.0 m.

Coefficient of permeability of material of the dam = 4×10^{-4} cm/sec

From the flow net for the dam (homogeneous in section) the following information was available :

Total number of flow channels = 5

Total number of potential drops = 12

5. (a) An elementary profile of a concrete gravity dam of triangular section has maximum water level up to the top of the dam which is 50 m. high d/s slope has a better of 0.8 H : 1.0 V. There is no tail water. Analyze the dam section and determine :

Factor of safety against overturning

Factor of safety against sliding

Uplift pr. intensity coefficient, acting on 100% area = 0.5

Coeff. of friction between base and foundation = 0.67

(CO1 and CO2)

P. T. O.

OR

- (b) A homogeneous earthen dam 20.0 m high with top width 6.0 m, upstream slope = 4 : 1 and downstream slope = 3 : 1, retain water to a depth of 15.0 m. An inverted filter bed provided extends to 30.0 m from toe point of the dam. Locate the phreatic line from the dam section and estimate the seepage flow through the body of the dam. Assume coefficient of permeability of material of the dam = 5×10^{-4} cm/sec.

(CO1 and CO2)

Roll No.

TCE-602

B. TECH. (CE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

REINFORCED CEMENT CONCRETE—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) Use of IS codes and tables provided by the CE Department is permitted.

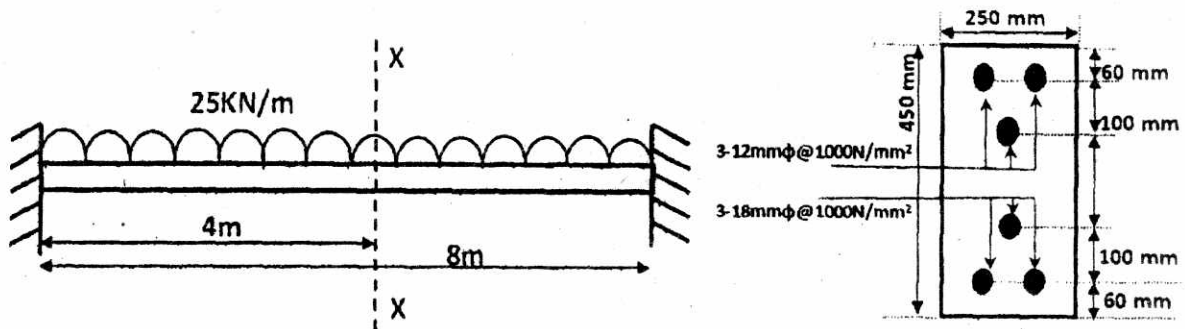
1. (a) Discuss the advantages and disadvantages of prestressed concrete members. (CO1)

OR

- (b) Review various prestress losses considered in PSC analysis, with their nature and significance. (CO1)

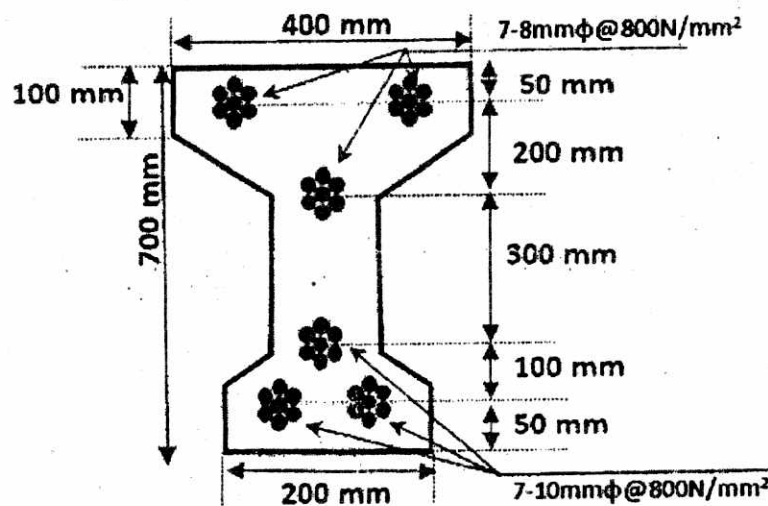
P. T. O.

2. (a) Determine the location of the thrusting force at the given pre-compressed beam section XX. (CO1)



OR

- (b) Inspect whether the given PSC beam section is concentric or not? (CO1)



3. (a) Derive formulae for calculation of any *three* types of prestress losses. (CO1)

OR

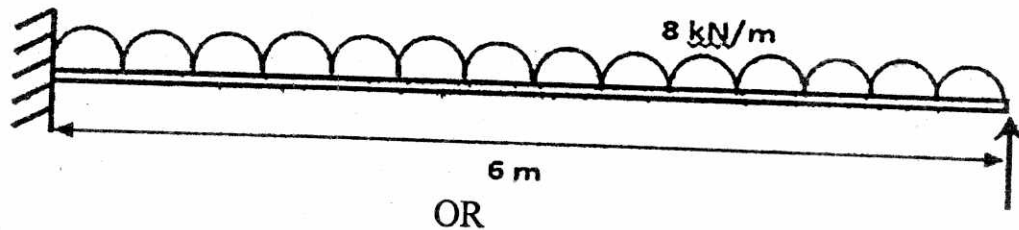
- (b) Discuss the steps and observations in developing a bending moment envelop for a statically indeterminate beam. (CO2)

(3)

4. (a) Illustrate the benefit of applying the limit analysis concept on a Propped Cantilever beam carrying UDL over the entire span. (CO2)

OR

- (b) Derive the formulae to determinate the shear force, bending moment and torsional moments in a beam circular in plan. (CO2)
5. (a) Construct the bending moment envelop for the given propped cantilever beam after 15% of moment redistribution. (CO2)



OR

- (b) Estimate the shift of point of contraflexure in the given fixed beam after applying 20% moment redistribution. (CO2)

